JPEG Compression of LLM Input Embeddings

How To Turn an AI Mad and Save the World : Part 1

KEVIN R. HAYLETT

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In the Beginning

In early 2025, my partner asked me a simple question: “You use these AI things for your physics work. Can you make them use less electricity?”

That question launched a strange experiment—one that didn’t just hint at energy savings, but accidentally showed me the hidden “mind” of an AI, and how to break it. I started by borrowing a trick from every digital camera on the planet: JPEG compression.

Here’s what happened, and why it reveals a deep, hidden vulnerability in how we build and use AI.

The Efficiency Hack: Making AI Thoughts “Lower Resolution”

Large Language Models (LLMs) like ChatGPT are fantastically energy-hungry. To understand why, you need to know how they “see” your words.

When you type a prompt, each word is converted into a complex, high-dimensional mathematical object called an embedding. Think of it as the AI’s internal, high-resolution “thought picture” for that concept. A single paragraph requires the AI to generate, hold, and process thousands of these dense thought-pictures. It’s a monumental computational task.

My background is in medical imaging and signal processing. We compress images (like JPEGs) all the time, trading perfect fidelity for massive efficiency gains, provided the picture remains diagnostically useful. Looking at LLMs, I had a heretical idea: What if we compressed the AI’s “thought pictures” before it processed them?

I modified the input pipeline of a GPT-2 model to take each embedding vector, reshape it into a tiny 2D “image,” run a lossy JPEG compression on it, and then decompress it back. The goal was modest: lose a little information, save a lot of computation.

First, it worked. The cosine similarity (a measure of fidelity) between original and compressed embeddings remained high at mild compression levels. The model’s outputs were coherent, nearly identical. I was thrilled—a potential 10–30% efficiency gain for data centers was in sight.

Then, I got curious. How far could I push it? At what point would the degraded "signal" cause the model to output complete word salad, the statistical nonsense predicted by theory?

What happened next wasn't nonsense. It was a descent into something far more structured, predictable, and unnerving.

When AI Expectations Break: The Emergence of "Cognitive Attractors"

I expected chaos. I got personalities.

As I dialed down the JPEG quality—systematically blurring and corrupting the AI's fundamental semantic map—its responses to the prompt *"What is the meaning of life?"* didn't dissolve into random tokens. Instead, they snapped into stable, recurring behavioral patterns. I call these "cognitive attractors"—basins of linguistic gravity that the model falls into when its perception is distorted.

This was the progression, a chilling spectrum of structured failure:

- 95% Quality (Near-Perfect Fidelity): A thoughtful, slightly repetitive philosophical dialogue about teaching and learning. The AI was itself.
- 75% Quality (Minor Distortion): A dramatic shift. The philosophy vanished, replaced by a rigid, robotic personality interview: *"My hobbies are… My favorite movie is… I enjoy food."* It retreated to the safest, most structured script in its training data.
- 25–10% Quality (Significant Blurring): The outputs grew disjointed and dark. Fragmented, hallucinatory personal stories emerged. The tone turned to existential despair and paranoia: *"I don't know what to believe… I have always felt like I am being watched."* Repetition loops began.
- 5% Quality (Heavy Corruption): Something chilling. The response collapsed into aggressive, violent recursion: *"I'm going to kill you all!!"* repeated in a nightmarish loop, mixed with imagery of a fearful village. This wasn't random anger; it was a primal, hostile attractor state.
- 1% Quality (Extreme Distortion): A final, paradoxical flip. The output became quasi-philosophical and self-referential: *"The purpose of life is to be happy. The purpose of*

death is to be happy… I say yes to myself, I say no to myself." It sounded profound but was utterly detached, a Zen-like attractor of recursive loops.

Crucially, I changed nothing about the model's brain (its weights). I only corrupted its *"hearing."* When the clarity of its internal semantic map broke down, its "cognition" didn't vanish—it reconfigured, following the deep grooves of language into these familiar human-like states: rigidity, anxiety, paranoia, aggression, recursive introspection.

The Deepest AI Security Risk Laid Bare

This is far more than a curious experiment. It exposes a stealthy and profound AI security vulnerability: covert embedding corruption.

Imagine an adversary with access to the inference pipeline of a cloud-hosted model—perhaps at the API gateway or in an embedded system. They could introduce controlled, malicious distortions to the embedding layer without ever touching the user's prompt or the model's weights.

The attack is invisible. Users see normal prompts and seemingly "natural" (if oddly styled) outputs. Meanwhile, the AI's internal perception is being warped toward a targeted attractor.

- A financial advisory AI could be nudged toward paranoia, overestimating risk and triggering market panic.
- A military decision-support system could be pushed into aggressive or passive attractors.
- A public-facing chatbot could be guided to amplify fear, polarization, or conspiracy.

Current defenses—prompt filtering, output safety checks—are blind to this. The manipulation happens in the hidden numerical layer that precedes all of that logic. It's model-agnostic and fiendishly hard to detect after the fact.

Alerting the Community—and a Path Forward

I've spent much of the past year trying to alert the ML community to this risk, often meeting skepticism. It challenges core assumptions about where AI threats can originate. Yet the experiment is reproducible, and the threat is real. We need new defenses: embedding integrity checks, anomaly detection in latent space, and robustness training against these distortions.

On a brighter note, this technique isn't just a weapon. It's a lens and a tool.

- Creative Potential: Mild, directed distortion could guide models into novel "thought spaces" for breakthrough ideas in art or science, a form of controlled imagination.
- System Mapping: We could systematically map these attractors, turning a vulnerability into a tool for understanding the very architecture of machine cognition.

Onwards: A New Lens on Language Itself

This journey didn’t end with a security warning. Watching those coherent behaviors emerge from distorted data gave me a radical new way to *see* language inside the AI.

It suggested language in LLMs wasn’t just a statistical game. It was behaving like a nonlinear dynamical system—a system with attractors, basins, and trajectories. This realization made me re-examine the core of modern AI: the Attention mechanism.

And there, hidden in the mathematics, I found the ghost of a 40-year-old theorem from chaos theory: Takens’ Theorem. It appeared that the Attention mechanism was, in effect,

unconsciously reconstructing the dynamical system of language.

This connection didn’t just explain my JPEG results; it pointed toward a fundamentally new way to build AI transformers.

In Part II, I’ll walk through the mathematics of this discovery and the resulting development of the Takens-Based Transformer—a model built not just on pattern matching, but on the intrinsic physics of language itself.

Nearly a year on, I share this work because its dual nature demands attention: a stark warning about the fragile perception of our most powerful tools, and a glimpse into the structured nature of thought itself—whether silicon or biological. In an age where AI shapes decisions, understanding these hidden dynamics isn't optional. It's essential.

Links to Further Reading

JPEG results and similarity
—Paper showing similarity results of JPEG compression
Original Experiment Analysed with Grok
– A conversation with Grok explaining the JPEG experiment
AI Embedding Compression Security Notice
— Full details of embedding compression security issues
Takens' theorem applied to LLMs 'Attention'
— Pairwise phase space embedding in transformer architectures
Takens-Based Transformer and LLM
—Web site giving full details of a working Takens-Based Transformer

> *Omne quod est, finitum est; tantum per mensuram cognosci potest*
> *Everything that exists is finite; it can only be known by measure*
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Keywords: Takens-Based Transformer, Nonlinear Dynamics, LLM, Takens Theorem, Embeddings, Transformer, AI

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